Performance (ex. PER)

Exercises Computer Architecture

1 Processor Benchmark & Performance

1.1 Which of the following statements are correct?

- ☐ The wall clock time is the total elapsed time , including I/O, Operating system overhead etc.
- ☐ Multi-threading improves the throughput of a process
- ☐ The CPU time does not include the I/O time
- ☐ Multi-threading improves the execution time of a process

1.2 What is the throughput?

- ☐ performance per Watt (the number of FLOPS per Watt)
- ☐ rate of processing work (n jobs/second)
- ☐ the time between start and completion of event/task/program (n seconds)
- ☐ the percentage of time a system is up and running

1.3 What is the SPEC?

- ☐ is a benchmark suit developed to measue performance based on the latest Java application feature
- ☐ is a benchmark that evaluates the power and performance characteristics of volume server class computers
- ☐ is the wordwide standard for measuring graphics performance based on professional applications
- ☐ is a bechmark suit designed to provide performance measurements that can be used to compare computer-intensive workloads on different computer systems

1.4 What is the goal of the EEMBC Benchmark?

- ☐ to evaluate performance of embedded microprocessors
- ☐ to evaluate integer computation performance
- ☐ to measure the energy efficiency of different computer systems
- ☐ to evaluate floating point performance

1.5 Which of the following is an energy efficiency metric?

- ☐ flops
- ☐ MIPS
- ☐ Performance per watt
- ☐ Power consumption

1.6 Both power consumption and performance per watt matters for an embedded system.

- ☐ True
- ☐ False

1.7 Processor performance

A program consists of 5'000 floating point instructions and 25'000 integer instructions. Processor A has a clock rate of $2.0GHz$. Floating point instructions take 7 cycles and integer instructions take 1 cycle.

a) How long does it take for this processor to run the program?

b) What is the average CPI for this processor for the given program?

c) Processor A runs program 2 consisting of 100'000 floating point instructions and 50'000 integer instructions. What is the average CPI for this program?

d) Processor B has an average CPI for program 2 of 3.5. It's clock rate is $1.8GHz$. How much time does it take to execute the program?

e) Which processor is faster and by how much?

Processor _____ is _____ times faster than processor _____.

1.8 Processor performance

Consider the following two machine designs with their respective CPI's for various instruction types. Computer A and Computer B have the same instruction set:

Instruction Type	CPI_A	CPI_B	Compiler 1 Mix
Data Manipulation	1.5	1.0	25%
Arithmetic	1.0	1.5	30%
Shifting	1.0	1.2	10%
Branching	4.0	2.0	25%
Multiply	20	12	10%

a) What is the average CPI for each of the computers using this program?

b) Computer A has a clock cycle time of $0.5ns$. Computer B is running at $1.8GHz$. Write a quantitative statement comparing the two computers.

c) What would the clock rate of the slower computer have to be to equal performaner of the faster computer?

1.9 Processor performance

A CPU run on a base frequency of $2GHz$. It executes a program with 5 million instructions with the given instruction mix.
What is the execution time of the program?

Instruction	Frequency	CPI_{instr}
ALU	50%	3
Load	20%	5
Store	10%	4
Branch	20%	3

1.10 Processor performance

A CPU is designed for an optimal performance on a given program with the following characteristics. 25% of all instructions are floating point instructions with an average CPI of 4.0, in addition the program contains 2% **F**loating **P**oint **SQ**uart **R**oot (FPSQR) instructions with an average CPI of 20. All other instructions have an average CPI or 1.33.

There are two design alternatives:

1. Decrease CPI of FPSQR instructions to 2.0
2. Decrease the average CPI of all floating point instructions to 2.5

Which choice is best?

1.11 Processor performance

We want to buy a new computer. It will mostly run programs P_1 and P_2.

What weigth w_1 and w_2 need the programs to have so that:

- CPU A the best buy?
- CPU B the best buy?
- CPU C the best buy?

Program	CPU_A	CPU_B	CPU_C
Program P_1 (sec)	1	10	100
Program P_2 (sec)	100	10	1

1.12 Processor performance

Given the following performance of two programs on three CPUs, user the geometric mean to calculate which computer is the fastest:

a) CPU A is the fastest?

b) CPU B is the fastest?

c) CPU C is the fastest?

Program	CPU_A	CPU_B	CPU_C
P_1 (sec)	40	15	20
P_2 (sec)	40	1000	150

1.13 Processor performance

Calculate the average CPI and the execution time for 5million instructions of the following instruction frequencies:

Instruction	Frequency	CPI_{instr}
ALU	40%	4
Load	30%	6
Store	5%	5
Branch	25%	4

1.14 What is the best metric for comparinc performance?

- ☐ arithmetic mean
- ☐ geometric mean
- ☐ median
- ☐ maximum performance
- ☐ harmonic mean

1.15 Processor performance

Calculate the execution time in ms, supposing to have CPU with the following instruction frequencies:

Instruction	Frequency	CPI_{instr}
ALU	45%	5
Load	25%	6
Store	10%	5
Branch	20%	3

For 2Million instructions and a CPU frequency of 3GHz.

1.16 Amdahl's Law

An improvement of the floating point execution unit resulted in 2x faster floating point instructions. In average 10% of all instructions are floating point instruction for this processor.

What will be the overall speedup?

1.17 Amdahl's Law

We want an overall speedup of 2 and can accelerate the floating point instructions by 4 times.

What should be the fraction of floating point instructions?

1.18 Amdahl's Law

A program consists of 2 different elements. Part A has a duration of 15 and part B a duration of 5 time units. There are two optimization variants:

1. optimization of the A part by two times
2. optimization of the B part by five times

Which optimization is more advantageous? What are the implications?